Q.3)

```
1111        for (int i=1 ; i<=n ; i++){
2222          for (int j = 1 ; j <= n ; j++) {
3333            Sop (i) ;
4444        }
            }
```

Time complexity :- $O(n^2)$

Space complexity :- $O(1)$

1) int i = 1
2) cond i<4
3) print (i)
4) increment (i++)

j<=1
1111
1111
1111
(4) (n) x(n) = (n^2)

Q.4)

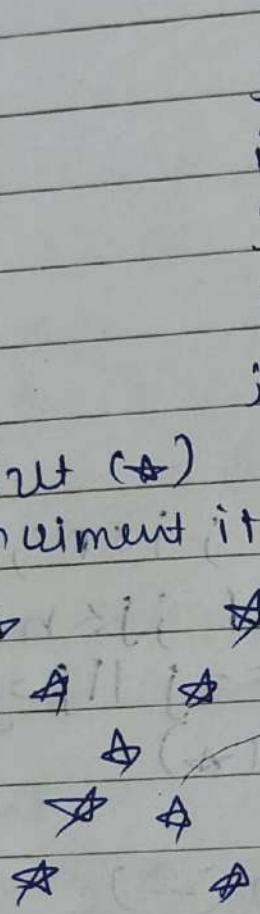
```
* * * * *      for (int i=1 ; i<=n ; i++){
*       *        for (int j=1 ; j<=n ; j++){
*       *          if (i==1 || i==n ||
*       *             j==1 || j==n)
* * * * *          {
                     Sop (*) ;
                   else
                     sop (" ")
                   }
```

Time complexity :- $O(n^2)$

Space complexity :- $O(1)$

1) int i = 1
2) condition i = 1 $\boxed{n}$ ×
j = 1, j = 2, j = 3, j = 4, j = 5 $\boxed{n}$
i = 2
j = 1 & j = 5 $n \times n = n^2$
i = 3
j = 1 & j = 5
i = 4 ~~& j =~~
j = 1 & j = 5
i = 5
j = 1, j = 2, j = 3, j = 4, j = 5
3) print (☆)
4) increment i++

Q.5]



```
for (int i=1 ; i<n ; i++){
    for (int j=1 ; j<n ; j++){
        if (i==j) || i+j==(n+1)){
            sop("☆").
        else
            sop(—)
```

Time complexity ⇒ $O(n^2)$
space complexity ⇒ $O(1)$

1) int i = 1
2) condition i = 1 $\boxed{n}$ $\boxed{n \times n = n^2}$
j = 1 and j = 5 $\boxed{n}$

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* $i=2$ n × $\boxed{n^2}$
$j=2$ & $j=4$ n
$i=3$ n × $\boxed{n^2}$
$j=3$
$i=4$
$j=2$ & $j=4$
$i=5$
$j=1$ & $j=5$
3) Print *
4) i++ -

Q.6] *
* *
* * *
* * * *
* * * *

```
for (int i=1 ; i≤n ; i++){
    for (int j=1 ; j≤n ; j++){
        if ( i==j || i>j)
            sop (*);
        else
            sop (--);
    }
```

Time complexity :- $O(n^2)$
Space complexity :- $O(1)$

1) int i=1 $\boxed{n}$
2) condition j=1 $\boxed{n}$ × n × n = $\boxed{n^2}$
i=2
j=1, j=2
i=3

1
2
3
4
5
⋮

j = 1, j = 2, j = 3.
i = 4
j = 1, j = 2, j = 3, j = 4
i = 5
j = 1, j = 2, j = 3, j = 4
3) ~~condi~~ print *
4) increment i++.

Q.7]

```
* * * * *
* * * *
* * *
* *
*
```

```
for (int i = 1 ; i <= n ; i++){
    for (int j = 1 ; j < n ; j++){
        if ( i+j <= n+1 ){
            cop (*);
        else
            cop ( - ) ;
```

Time complexity = $O(n^2)$
Space complexity. = $O(1)$

1) int i = 1 $\boxed{1}$ × $n \times n$
2) condition j = 1, j = 2, j = 3, j = 4, j = 5 $\boxed{n}$ $\boxed{n^2}$
   i = 2
   j = 1, j = 2, j = 3, j = 4,
   i = 3
   j = 1, j = 2, j = 3
   i = 4
   j = ~~1~~, j2
   i = 5
   j = 1
3) ~~cou~~ print *
4) Increment

Q.8) ☆ # ☆ # ☆
☆ # ☆
☆ # ☆ # ☆
☆ # ☆
☆ # ☆ # ☆

```
for (int i = 1; i ≤ n; i++)
for (int j = 1; j ≤ n; j++){
~~sop(#~~ if (i+j) % 2 == 0
        sop(#);
    else {
        sop(☆);
    }
```

Time complexity $= O(n^2)$
Space complexity $= O(1)$

1) int i = 1 j ≤ n if i+j % 2 ==
2) condition, j = 1, j = 3, j = 5
int ~~else~~ i = 2 j = 2, j = 4
~~#~~ i = 3 j
j = 1, j = 3, j = 5
i = 4
j = 2, j = 4
i = 5
j = 1, j = 3, j = 5
else
i = 1, j = 2, j = 4
i = 2, j = 1, j = 3, j = 5
i = 3, j = 2, j = 4
i = 4, j = 1, j = 3, j = 5
i = 5, j = 2, j = 4
4) print ☆ and #
5) increment i++.

Q.9)

```
1                for(int i=1 ; i<=n ; i++)
12               for(int j=1; j<=n-i ; j++){
123                  cop("-");
1234             }
123              for (int j=1 ; j<=(2*i)-i ; j++){
12                   cop(j + " ");
1                }
                 }
                 for (int i=n-1 ; i>=1 ; i--){
                     for (int j=1 ; j<=n-i ; j++){
                         cop(" ");
                     }
                     for (int j=1 ; j<=(2*i)-1 ; j++){
                         cop(j);
                     }
                 }
```

Time complexity $O(n^2)$

Space Complexity $O(1)$

i) int i=1

2) condition $i \le n$

for space $n-i$, $4-1 = 3$

$i=1$, $j=1$, $j=2$, $j=3$, 1

$i=2$, $j=1$, $j=2$, 2

$i=3$, $j=1$ (n) 3 $= (4)$

$i=4$, $j=0$ 4 $= (n)$

For print j = $i=1$, $j=4$, 1

$i=2$, $j=3$, $j=4$ 2 $=(4)$

$i=3$, $j=2$, 3, 4 3 $=(n)$

$i=4$, $j=1$, 2, 3 4

for print 2xi

$n-i = 4-3 = 1$

$i = n-1$

$i = 4-1 = 3$

for space = $i \le 3$

(3) $i=3$, $j=1$, 1

(n) $i=2$, $j=1$, $j=2$, 1

$i=1$, $j=1$, $j=2$, $j=3$

n^2+n $((n) \times (n) + n)$

$4n^2 + 2nn^2 + n + (n) \times (n) + n)$

$\boxed{n^2}$

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10)

```
1                1
12             21
123          321
1234 4321
1234 4321
123          321
12             21
1                1
```

```
n for (int i=1; i<n; i++){
    n for (int j=1; j<i; j++){
        sop (j)
    for (int j=1; j<2*(n-i); j
        sop ( ) }
    for (int j=1; j>i; j--){
        sop (j).
    }
    for (int i=n; i>1; i--){
        for (int j=1; j<i; j++){
            sop (j)
    }
    for (int j=1; j<2(n-i); j++)
        sop (j).
    for (int j=i; j>1; j--){
        sop (j-1);
```

$n^2+n^2+n^2+n^2 = 4n^2$

$n \times n = n^2 +$

$n \times n = n^2 +$

$n \times n = n^2 +$

$n * n = n^2$

Time complexity :- $\boxed{O(n^2)}$

Space complexity :- $\boxed{(0+O(1))}$

1) int i=1 ~~space~~

2) cond $j \le n$ j=

j=1 , (n) *

i=2, j=1, j=2

i=3, j=1, j=2, j=3

i=4, j=1, j=2, j=3, j=4

space $2(n-i)$ $2(4-1)$

$2 \times 3 = 6$

i=1, j=2, j=3, j=4, j=5, j=6, j=

i=2, j=3, j=4, j=5, j=6 1

i=3 j=4, j=5 (n) (3)

i=4 ~~j=0~~

j= (4)

i=1, j=8 1 (n)

i=2, j=7, 8 1

i=3, j=6, 7, 8 1

i=4, j=5, 6, 7, 8 1

11)

```
1   1
2   2
3
4   4
5   5
```

```
for (int i = 1 ; i<n ; i++){
    for (int j = 1 ; j<n ; j++){
        if ( i==j || j==n-i+1)
            sop (i);
        else
            sop ( );
```

Time complexity → $O(n^2)$

Space complexity → $O(1)$

1) int i=1 n n

1,2) condition i=1 ✗, j=1, j=5 (1) n ✗

i=2 ✗, j=2, j=4 (11) n ✗

i=3, j=3 (1) n

i=4, j=2, j=4 n

i=5, j=1, j=5 n

$= O(n^2)$

3) print j

4) Increment i++

12)

```
1
2 3
4 5 6
7 8 9 10
11 12 13
14 15
16
```

```
for (int i = 1 ; i < n ; i++) {
for (int j = 1 ; j < n-i ; j++) {
    } sop (-);
for (int j = 1 ; j < (2*i) - i ; j++
    } sop (j);
for (i = n-i ; i >= 1 ; i--) {
for (j = 1 ; j <= n-i ; j++) {
    sop (-);
    } for (j = 1 ; j <= 2 x (n-i) + 1 ; j++) {
        sop (j);
    }
```

Time complexity :- $O(n^2)$

space complexity :- $O(1)$

1) int $i = 1$ $(n-i) = 3$

2) cond $i = 1$, $j = 1, 2, 3$

$i = 2$ $j = 1, 2$

$i = 3$ $j = 1$ space

$i = 4$ $j = 0$ (n)

print $j = 2 \times 1 - 1$

$i = 1$, $j = 4$

$i = 2$, $j = 3, 4$

$i = 3$, $j = 2, 3, 4$

$j = 4$ $j = 1, 2, 3, 4$

$2 \times (4-3) + 1$

$i = n - 1 = 3$ (n)

$i = 3$ $j = 1, \cancel{j = 3, j = 4}$

$i = 2$ $j = 1, 2$

$i = 1$ $j = 1, 2, 3$

$i = 3$ $2 \times (n-i) + 1$ $2+1$

$j = 2, 3, 4$ (n)

$j = 2$ $j = 3, 4$

$i = 1$, $j = 4$

$(n) \times (n) + (n) + (n) \times (n) + (n)$

$n^2 + n^2$

$n^2 + n^2$

$2n^2 + 2n^2$

$4n^2 = (n^2)$

13)

```
1
01
101
0101
101
01
1
```

```
for (int i=1 ; i≤n; i++){
    for (int j=1 ; j≤n-i ; j++){
        sop (-)
    }
    for (int k=1 ; k≤i ; k++){
        int num = i+k;
        if (num % 2 == 0)
            sop (1);
        else
            sop (0);
    for (int i = n-1 ; i≥1 ; i--){
    for (int j = 1 ; j≤ n-i ; j++){
        sop ("-");
    }
    for (int k = 1 ; k≤(2×n)-i ; k++){
        int num = i+k;
        if (num % 2 == 0)
            sop (1)
        else
            (0);
```

Time complexity :- O(n)

Space complexity :- O (1)

1) int i=1 space

2) cond

$n-i = 4-1 = 3$	$j = n$.
i=1, j=1, j=2, j=3	i=1 ; j=4
i=2, j=1, j=2	i=2, j=3, j=4
i=3, j=1	i=3, j=2, j=3, j=4
i=4, j=0	i=4, j=1, 2, 3, 4

4-3

Space n-i

i=1, j=i

i=2, j=1, j=2 (n)

i=3, j=1, j=2, j=3

i=4

3) print j

4) increment i++

$(n) \times (n) + (n) + n$

$(n) \times (n) + (n)$

$(n^2 + n^2) + (n^2 + n^2)$

$2n^2 + 2n^2$

$4n^2$ (n^2)

j=

i=3 j=1, j=2, j=3

i=2 j=3, j=4 (n)

i=1 j=4

$(n^2 \times n) + n + (n \times n) + n$

$= n^2 + n^2 + n$

$= 3n^2 + 3n^2$

$= 6n^2$

$= \boxed{n^2}$